SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques

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using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I **Keerthana R (192524489)** (Name / Reg No) certify that this submission is my original work and*
that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Keeerthana R

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

IP Address: **192.168.10.65**
Subnet Mask: **255.255.255.192 (/26)**
Default Gateway: **192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

• Identify the subnet to which the configured IP address belongs.

• Determine the network address, broadcast address, and valid host range.

• Verify whether the configured IP address and gateway are valid.

• Identify the configuration errors.

• Optimize the configuration by assigning an appropriate IP address and default gateway.

• Calculate the total number of usable host addresses after correction.

ANSWER:

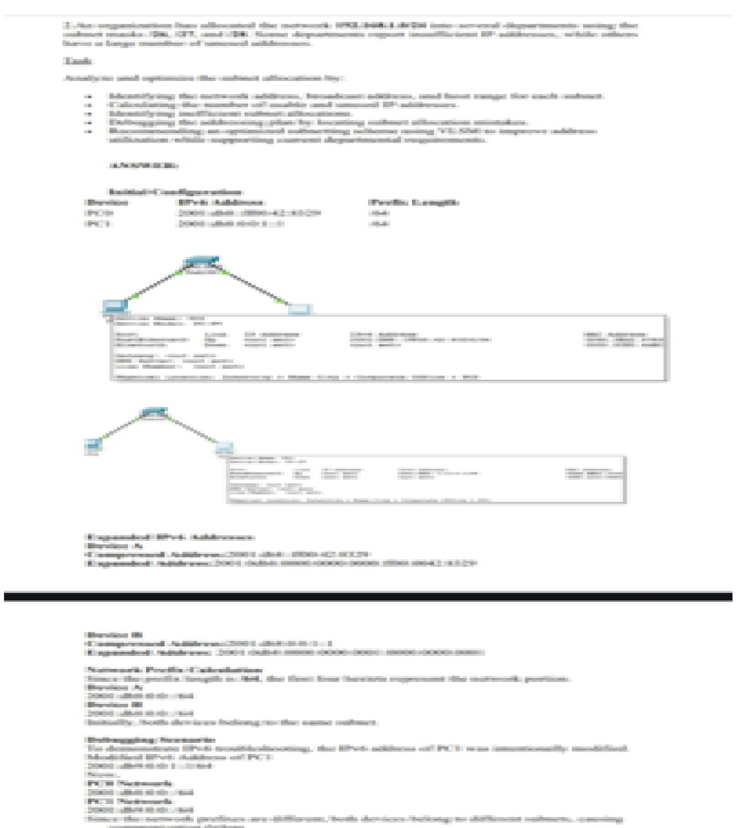
Initial Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.65	255.255.255.192 /26	192.168.10.1

Subnet Identification

A /26 subnet has a block size of 64 addresses. The configured IP **192.168.10.65** belongs to **192.168.10.64/26**, not 192.168.10.0/26.

Cisco Packet Tracer Output Image (as provided)



Network Address, Broadcast Address and Host Range

Parameter	Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126
Total Addresses	64
Usable Host Addresses	62

Verification of Configured IP and Gateway

• **192.168.10.65:** Valid. It is in the host range 192.168.10.65–192.168.10.126 of 192.168.10.64/26.
• **192.168.10.1:** Invalid as the gateway for this workstation. It belongs to 192.168.10.0/26 (host range 192.168.10.1–192.168.10.62).

Configuration Errors

1. The workstation is in **192.168.10.64/26**.
2. Gateway **192.168.10.1** is in **192.168.10.0/26**.
3. The workstation and gateway are in different subnets, causing communication failure.

Debugging Calculation

255.255.255.192 = /26 → 26 network bits + 6 host bits. Block size = 256 – 192 = **64**. Therefore 192.168.10.65 falls in 192.168.10.64/26.

Optimized Configuration

Assign a valid host address and a gateway from the same /26 subnet.

Device	Correct IP	Prefix / Mask	Gateway
PC0	192.168.10.66	/26 (255.255.255.192)	192.168.10.65

Why this configuration is correct

PC0 192.168.10.66 and gateway 192.168.10.65 both belong to 192.168.10.64/26. The gateway is a valid host address.

After Debugging – Expected Packet Tracer Test

```
C:\> ipconfig
IP Address........: 192.168.10.66
Subnet Mask.......: 255.255.255.192
Default Gateway...: 192.168.10.65

C:\> ping 192.168.10.20

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128 (4 replies)

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Total Number of Usable Host Addresses After Correction

Host bits = 32 – 26 = 6. Total addresses = 2^6 = 64. Usable host addresses = 64 – 2 = **62**.

The two reserved addresses are network 192.168.10.64 and broadcast 192.168.10.127.

Conclusion: 192.168.10.65 is a valid host address for 192.168.10.64/26, but 192.168.10.1 is an incorrect gateway because it belongs to 192.168.10.0/26. Using a gateway from the same subnet resolves the mismatch. The corrected /26 subnet provides **62 usable host addresses**.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated

Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation